Frugal Innovation: A Systematic Literature Review

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Mapping the frugal innovation phenomenon

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ABSTRACT

This study aims at mapping the frugal innovation phenomenon explore. The study reveals that scholars affiliated with Indian institutes and originated from India have played a key role in this research discipline. Country wise, the highest number of frugal innovation cases is from India. The articles on frugal innovation have published in a wide range of disciplines and journals. Scholars, practitioners, and policy-makers have understood frugal innovation concept in various ways. Studies are predominantly in sectors, such as healthcare, electric and electronics, transport, finance, ICT, and energy. On the contrary, despite high importance, agriculture and education sectors have received limited attention.

1. Introduction

Frugal innovation has emerged as an important concept for scholars, practitioners and policy makers. It is highly relevant for multinationals, small and medium enterprises (SMEs), non-government organizations, and state organizations [28,83]. Until recently, frugal innovation was an unknown phenomenon. Frugal innovation is considered as a new source of innovation mainly to meet the needs of low-income customers. Hence, it has primarily been explored emphasizing on affordability. However, what is frugal innovation is still under a great debate [101]. There are numerous definitions of frugal innovation [14,43,75]. Hossain et al.([43]: 133) define frugal innovation… "as a product, service or a solution that emerges despite financial, human, technological and other resource constraints, and where the final outcome is less pricey than competitive offerings (if available) and which meets the needs of those customers who otherwise remain un-served".

Misconceptions and multiple conceptions are associated with the concept [71,75,101]. The frugal innovation concept overlaps with over a dozen of other concepts including cost, good-enough, the base of the pyramid, inclusive, grassroots, disruptive, jugaad, and reverse innovations [71,75,101]. All of these concepts have a common denominator: developing low-cost but good enough product, service, and business model for low-income customers in developing countries. However, reverse innovation concept goes a step further. It encompasses some set of customers in rich countries. Developing capability of frugal innovation is a prerequisite for reverse innovation [102]. When an innovation is used in the developing countries and then comes to the developed countries, it is considered as reverse innovation [33,43]. The overlap of frugal innovation with some other concepts hinders the development of literature on it [71]. Some recent studies have explored how frugal innovation concept is different from the overlapping concepts [2,71,101].

Even though frugal innovation is highly cited in the popular press, there are limited scholarly publications on this phenomenon [45]. Most of the studies on frugal innovation are conceptual or case-based. Some prominent frugal innovation cases have been frequently cited in the literature. What cases are considered as frugal innovation is not clear. Hence, developing a comprehensive list of frugal innovation cases, their sectors, and their origins, etc. is crucial so that scholars, practitioners, and policy makers get an overall understating of the phenomenon. This study aims at mapping the frugal innovation phenomenon. This study reveals the popular outlets of publications on frugal innovation, most frequently used keywords in the studies, dominant sectors, countries of origin, and a comprehensive list of frugal innovation cases.

The remainder of the paper is structured as follows. The following section presents the current understanding of the frugal innovation phenomenon. Section three explains the methodological approach of this study. Section four includes the findings of the study. Implications, limitations, and future research avenues are presented in the final section.

2. Current knowledge on frugal innovation

The number and quality of frugal innovations are increasing dramatically [83,100]. Frugal innovations generate better business and social value than traditional innovations. Angot & Plé [7] argue that frugal innovation includes four main characteristics: affordability, good performance, sustainability, and usability.Hartley [37] provides some propositions intending to pursue academics, managers, and policy-makers to consider new approaches to innovation. Gewald et al. [31] advocate that firms should develop frugal innovations considering the needs of poor customers as the starting point and work back-

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ward by optimizing products features down to essentials. Bhatti & Ventresca [12,13] emphasized that the frugal aspect involves solving problems without being stymied by affordability, resource, and institutional constraints.

Frugal innovations are cheap, robust in harsh environments, easy to use and repair, new uses of existing technologies, and made of used and local materials [25,78]. According to [54]; frugal innovation improves the well-being of poor people in four dimensions: (1) income generation and security, (2) education, (3) infrastructure, and (4) distribution. It reduces the necessity of sophisticated labs, instead relies on basic engineering skills [59]. Moreover, Prabhu & Gupta [70] argue that three broad innovation heuristics are used for frugal service innovation: (1) combining existing materials, processes and resources through bricolage, (2) reducing time, materials and human resources, and (3) creating self-service options for users. However, Lazcano [55] believes that a global business continuity program is necessary to foster frugal innovation.

The developing countries are increasingly embracing innovation in their national innovation policy [10,23,98]. They have prioritized their agenda to boost frugal innovation [16]. Recently, planning commission of India has emphasized on frugal innovation for inclusive growth [9,63,71]. Even though income level of many customers are growing, they have still little access to many essential needs [101]. Mukerjee [62] believes that affordability is a key to serving low-income customers with frugal innovation. On the contrary, Gupta [36] argues that frugal innovation is beyond affordability because some aspects, such as collecting sachet for shampoo and hair oil are expensive. Although frugal innovation has primarily been explored emphasizing on affordability, even low-income customers are increasingly looking for attractive designs of products [92,93].

There has been a substantial shift in global innovation and Western MNCs are increasingly preferring countries, such as China (62%) and India (29%) as a source of innovation [95]. Thus, the technical and social changes in the developing countries seem to become key drivers of innovation in near future [68]. By exploring subsidiaries of Bosch and 3M in India, Ojha [66] found that these MNCs are reframing their global strategies and emphasizing more on low-income countries to develop frugal innovations. Most sophisticated frugal innovations are developed by research and development (R&D) subsidiaries of Western firms in emerging countries with high autonomy [100]. MNCs such as General Electric (GE) and Siemens are continuously bringing frugal innovation into the market to tap new customers [47,78].

Frugal innovations in emerging market take place in three levels [52]. At the first level, many Western MNCs now have full-fledged R&D centers in emerging countries, such as India and China. Among fortune 500 listed firms, 98 have R&D facilities in China and 63 in India [28,81]. At the next level, local MNCs pursue innovative research in their home and other emerging countries. Multinationals of emerging markets, such as India and China are developing inexpensive products considering the resource-constrained environment and low-income customers. Their products provide high value per price. Tata Nano – the cheapest car in the world, and Tata Swach – water purifier by Indian conglomerate Tata have provided a new way of serving underserved customers in the developing countries.

At the third level, innovations are emerging from the grassroots level. Low- or not-educated grassroots innovators with their frugal mindset develop products and services aiming to meet local needs. For example, Mitticool – a clay fridge developed by a potter, and milking machine by a school teacher both in India are prime examples of how these frugal innovations are meeting local needs. India is believed as the epicenter of frugal innovation [63,78]. However, Prathap [72] debunks that belief and argues that other countries are more active in frugal innovation than India. Frugal innovations, such as Embrace and mOm by Western individuals are available at 10–20% (several hundred dollars) of the price of traditional baby warming products. Many non-government organizations are also playing a key role in frugal (service) innovations [80]. Thus, frugal innovations are emerging from a wide range of sources.

The developing countries possess some competitive advantage over the developed countries, such as skilled human capital at a very low cost and a mega market for products [30]. Many frugal innovations trickle up Western countries with some modifications to satisfy Western customers [43]. Frugal innovation is not all about the product rather it entails services, too (Williamson and Zeng, 2009). Western MNCs are developing products in co-creating with customers, empowering local engineers with a clean slate approach for product development in the developing countries [50]. The economic state of the developed countries is plummeting. For example, the median income of US household decreased by 7.5% from 2007 to 2014 [24]. Gross disposable income of households in most European countries has also decreased. Therefore, many customers in the developed countries are increasingly becoming poorer. Consequently, they look for affordable products to meet their tight budgets. Firms need a novel strategy to sell products that are within the purchasing power of low-income customers [7]. Therefore, embracing frugal innovation is important to meet the needs of some sets of customers in the developing and developed countries.

3. Research methods

3.1. Article searching process

To search articles on frugal innovation, we adopted a systematic approach (see Tranfield et al., 2003). Fig. 1 depicts the steps of the article (including book chapters and working papers) searching process (N indicates the number of articles). We used "frugal innovation" as a searching term on five databases. Only articles in English were included in searching. The first attempt of searching was on the Web of Science and 24 articles were found from there. To include more articles, the second attempt was made on the Scopus database and 55 articles were retrieved from there. Removing 40 overlaps (and irrelevant articles) between the Web of Science and the Scopus, 39 articles were considered from those two databases. The third attempt was made on the EBSCO database using academic search elite and 50 articles were identified. After removing overlaps, 42 articles were found from the EBSCO database. Thus, we found 81 articles altogether from the Web of Science, Scopus, and EBSCO. The fourth attempt is made on the Google scholar and 10 more articles were found. Finally, particularly to include more recent articles, we looked at the Social Science Research Network (SSRN) database where many forthcoming articles and working papers are included. However, only three articles were found from the SSRN. Thus, 94 articles were found from the five databases.

Altogether 94 articles were considered to quickly cheek if they really focus on frugal innovation. For the simplicity, the term article is used, which encompasses journal and conference articles, book chapters, and working papers. Articles, which just merely mentioned frugal innovation without really focusing on frugal innovation, were removed from the list of articles. Finally, 62 articles were considered for this study purpose.

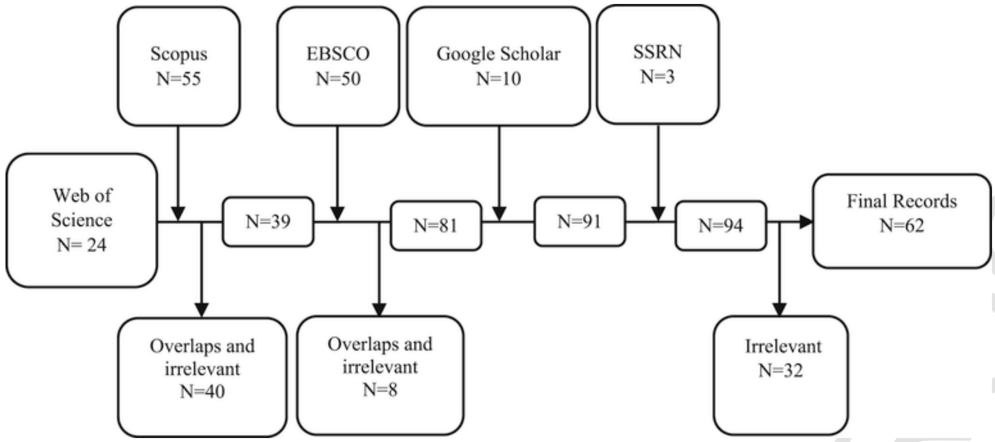

Fig. 1. Process and steps of searching relevant article.

3.2. Retrieving information from the articles

The following information, among others, is collected and tabulated for review analysis: case name, the sector of cases, developer, country of origin, authors by country, the name of the journal or other outlets, year of publication, and what problem the case solves. The frequency of cases is calculated. Thus, a list of frequency of frugal innovation cases has been made. However, an important note is that 13 articles did not mention any case of frugal innovation. Altogether, 117 cases of frugal innovation were found from the literature. The most frequently cited cases of frugal innovation are Tata Nano, GE's ECG machine, GE's ECG machine, Tata Swach, Aravind Eye Care, M-Pesa, Aakash, Vortex ATMs, and Narayana Hospital (see Appendix 1). For sectorial analysis, necessary information on each frugal innovation case was extracted from the selected articles and then from external sources to gain adequate knowledge of the cases.

4. Results

4.1. Descriptive analysis

This section is organized into two parts. The first part consists of descriptive statistics of the publications. The second part presents the sectorial analysis of frugal innovation studies and cases.

4.2. Authorship by country

Fig. 2 presents the frequency of appearance of authors by country. The authors of frugal innovation studies are from 18 countries (Fig. 2). Authors from India share much higher in authorship than authors of any other country. Authors of both UK and US have the same number of appearance. Most of the other authors are located in the developed countries with the exception of the following developing countries: Argentina, Jamaica, and South Africa.

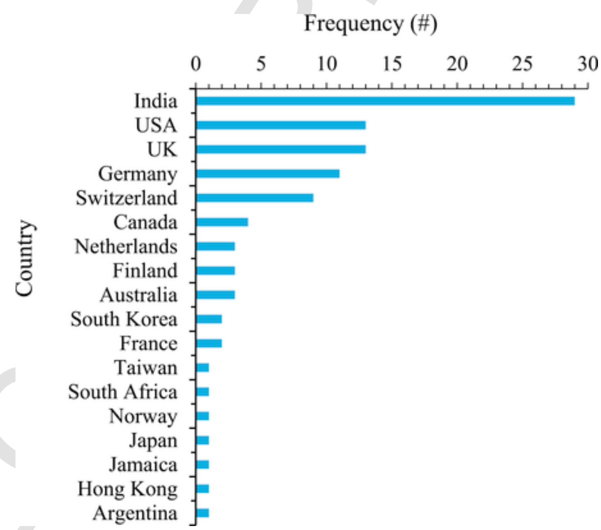

Fig. 2. Frequency of authorships by country.

4.3. Outlets of frugal innovation studies

Table 1 shows the publication of frugal innovation by discipline and journal. Journals include 30 articles and the rest of the articles have been published in another form of outlets. Studies on frugal innovation have appeared in eight major disciplines. Journals of Technology and Innovation category share the highest number of articles (36%) followed by management (20%) and health (13%). Moreover, business, ICT, marketing, strategy, and sustainability are other disciplines wherein articles on frugal innovation have been published. Altogether, 30 journal articles have appeared in 26 journals. Managerial journal Research-Technology Management comprises three articles followed by Globalization and Health and Journal of Indian Business Research each with two articles. The remaining 23 journals include one article each.

Table 2 depicts the keywords that have appeared most frequently in the literature. Altogether 243 occurrences of keywords found in the articles. Removing overlap of keywords, 133 distinct keywords have been found. As expected, frugal innovation has appeared the highest number of times. Most frequently used keywords are Emerging countries, Reverse innovation, Innovation, Base of the pyramid, and Disruptive innovation. The appearance of keywords, such as Affordability, Frugal, and Jugaad, Constraints, Cost innovation indicates that frugal innovation is highly related with the low-cost aspect. Frugal innovation is discussed with other concepts, such as Open innovation, Sustainability, Social innovation.

A complete list of cases has been presented in Appendix 1. Altogether, 117 frugal innovation cases were retrieved from the literature. Table 3 provides the distribution of 84 cases by sector. Altogether, detail information about 33 cases was not found from the literature because they were mentioned just by name with no description.

Table 1
Publications on frugal innovation by discipline and journal.

Journal by discipline	Frequency (#)
Technology and Innovation	11
Innovation and Development	1
International Journal of Business Innovation and Research	1
International Journal of Technology Management	1
Journal of Technology Management for Growing Economies	1
Research-Technology Management	3
South African Journal of Science	1
Technology in Society	1
Technology Review	1
Technovation	1
Management	6
Asian Journal of Management Cases	1
IIMB Management Review	1
International Journal of Rural Management	1
Management Research Review	1
Public Money & Management	1
Contemporary South Asia	1
Health	4
BMC Health Services Research	1
Globalization and Health	2
PLoS Medicine	1
Business	3
Ivey Business School	1
Journal of Indian Business Research	2
ICT	2
Communications of the ACM	1
Information Technology for Development	1
Marketing	2
Industrial Marketing Management	1
Journal of Strategic Marketing	1
Strategy	1
Journal of Business Strategy	1
Sustainability	1
Journal of Management for Global Sustainability	1
Total	30

Table 2
Most frequently appeared keywords in the articles.

Keyword	Frequency (#)
Frugal innovation	30
Emerging countries	11
Reverse innovation	10
Innovation	9
Base of pyramid	8
Disruptive innovation	7
Global automobile industry	7
India	6
Affordability	4
Frugal	4
Jugaad	4
Open innovation	4
Sustainability	4
Health care	3
Innovation management	3
Constraints	2
Cost innovation	2
Developing country	2
Institutional voids	2
IT capabilities	2
IT-enabled innovation	2
Lead Markets	2
New product development	2
Product Innovation	2
Social innovation	2

Moreover, the attempt to find information on those 33 cases from other possible sources has resulted in unsuccessful outcomes. Of the 84 cases, one-third of them is in the healthcare sector followed by, in descending order, electric and electronics, transport, finance, ICT, energy, house, and water. However, cases of electric and electronics can also be considered in various other sectors based on their applications. Table 4 shows that around 50% of cases are from India followed by China and USA. Less than one-third of the cases are from the countries other than India, China, and the USA.

Table 3
Frugal innovation cases by sector.

Sector	Case (#)	Frequency (%)
Healthcare	28	33
Electric and Electronics	13	15
Transport	8	10
Finance	8	10
ICT	7	8
Energy	5	6
Housing	4	5
Water	4	5
Education	2	2
Consumer goods	2	2
Agriculture	2	2
Aeronautic	1	1
Total	84	100

4.4. Sectoral analysis

This section presents the development of frugal innovation literature in various sectors. As mentioned earlier, an important note is that the cases of electric and electronics sector may fall in other sectors in terms of applications. Therefore, electric and electronics have not been considered as an independent sector in the following discussion.

4.5. Healthcare

Affordable healthcare has received enormous attentions from various agencies [90]. Some Western MNCs such as GE and Siemens are continuously developing frugal healthcare devices to meet the affordability and the local context of low-income customers. Pai et al. [67] argue that medical devices need to be designed in resources-constraints setting with the lens of frugal or reverse innovation to ensure that they are robust, appropriately priced, and tested in a variety of conditions. GE, for example, has developed $1000 handheld electrocardiogram (ECG) device, $15,000 ultrasound machine, and Lullaby baby warmer to provide high quality care for underserved customers [3,4,8,34,56,82]. Thus, an ECG test in India and other similar socio-economic countries costs US$1, which is 20% of the cost in the developed countries [7]. Siemens' Fetal Heart Rate Monitor (FHM), developed in India, is used globally as an inexpensive device to monitor the heart rate of fetuses in the womb. Embrace and mOm are two prominent portable infant warmers, developed by Western individuals, which have potential to save millions of babies who otherwise die [13,53]. Two anesthesiologists from the UK developed Shakerscope, a light source for a clinical examination for the eye, ear, and throat from their personal experiences in Zambia [59]. Shakerscope provides enough electricity for 3 min on shaking for 30 s [60].

An Indian grassroots innovator invented Jaipur foot, the low-cost artificial limb to meet the local needs [14]. Later, local doctors in col-

Table 4
Major sources of frugal innovations by country.

Country	Frequency (%)
India	48
China	12
USA	11
Others	29
Total	100

laboration with MIT developed Jaipur foot, which has been used in various countries [4]. Jaipur foot is a cheap, mass-produced rubber prosthesis suitable for uneven terrain in the developing countries, such as India [25]. Moreover, Jaipur remotion knee is made of titanium with price ranges of $5 to $20. It is simple, easy to manufacture, and performs high [14]. Many small firms from the developing countries are making frugal medical devices to meet their local needs [49].

Indian hospitals are increasingly serving their customers at an ultra-low cost. Aravind Eye Care Hospital and Narayana Hridulaya hospital in India are bringing novel ways of treatment with low-cost and cross-subsidy [34]. Aravind Eye Care Hospital in India, for example, provides 60% of their cataract surgeries free of cost from subsidized payments by patients who still pay only $30 for a surgery compared to $3000 in the USA [29]. Some developing countries, such as Ghana, Mali, Mexico, and South Africa are actively using mobile technologies to improve healthcare system [69]. Mexico's Medicall, for example, adopted a subscription model to offer professional healthcare advice at $5 monthly fee. mPedigree program in Ghana and SMS service in Mali allow patients to receive a prescription, medial test reports, and medicines from their homes. HealthLine Service in Bangladesh provides round the clock teleconference between a patient and a licensed physician. Some medical journals, Globalization and Health, are publishing papers focusing on frugal medical services [21]. Studies show that embracing open innovation – inflow and outflow of knowledge – can be useful to stimulate frugal and reverse innovations especially in the healthcare sector [22,44]. Thus, frugal innovation (device, service, process, business model) has emerged as a blessing to serve unserved patients.

4.6. Information and communication technology

Frugal innovation in the information and communication technology (ICT) sector is ubiquitous. ICT sector has experienced frugality with the development of hardware, software, and business models. Western MNCs are facing fierce competition and they need to provide more value for money. Nokia 1100 and Nokia 101 mobile handsets brought by Nokia Corporation had provided communication facilities for millions of low-income people in the developing countries [4,83]. Micromax is, an Indian consumer electronics firm, skyrocketing its growth in India. Its mobile phones do not need recharging for a month or longer. Its flagship phone with 30 days of standby charge is available at $45 [85]. An American firm Logitech needed to develop sophisticated mouse to compete with a Chinese local firm Rapoo [27]. One Laptop per Child PC ($100) has transformed how children in the developing countries can access to modern learning [13,51]. A US$60 dollar Aakash tablet was distributed to students throughout India by the Indian government [4,5]. Virtual Interactive Classroom in Bangladesh driven by a mobile e-learning environment shows how underprivileged children get access to quality education [74].

Mobile banking has become a new normal in many developing countries. In 2007, a Kenyan leading mobile network operator Safaricom launched mobile phone-based money transfer and microfinancing service M-Pesa that has revolutionized not only financial transactions but also communication patterns in Africa [4,46]. Eko in India, Easypaisa in Pakistan, Kopo Kopo in Kenya, WIZZIT in South Africa, and bKash in Bangladesh are enterprises that are active in mobile banking [12,13,94]. In India, mobile service provider Bharti Airtel developed a new business model to deliver the cheapest mobile talk time of the world and gained an impressive profit margin [16]. Access to mobile communication has changed the lives of millions of poor people in the developing countries. They can now get update information of market prices of their products in a fraction of time. Thus, they are better informed and have high bargaining power in trading. Microfinance lenders and borrowers track their loans and repayments via mobile technology. Mobile technologies are replacing the high street banks by providing an easy way to pay bills, tracking expenditure and wealth management via text messages [69]. Moreover, Barclay [11] points out that frugal innovation is useful to promote successful cybercrime legislative process. Thus, the combination of hardware, software, and new frugal approach is significantly improving the lives of underserved customers in the developing countries.

4.7. Transportation

The transport sector is changing very rapidly not only in the developing countries but also in the developed countries. Indian conglomerate Tata Motors introduced the cheapest car of the world called Tata Nano which was launched in India at a price tag of US$ 2000 in 2009 [79]. Tata Nano follows the frugal efforts [15]. It is an appropriate car for Indian and other similar markets for some key reasons: the rise of middle class in the developing countries, replacement of two-wheeler, lower expectation and little aware of standard features compared to Western customers, narrow road, high traffic, and lack of enforcement of proper government safety measures, among others [69]. The main features of Tata Nano include two-cylinder, 624 cc engine and lightweight (1300 pounds) [76]. Tata Nano is an unprecedented innovation in the automobile industry, which shows a path to building innovation capability that has remained largely unexplored [58]. Tata Ace is another frugal innovation, which is claimed to be India's first four-wheel mini truck launched in 2005 [90]. The cheapest version of the Dacia Logan car ($6000) is popular worldwide [13]. India's Mahindra and Mahindra posed a severe threat to the USA firms in small tractors segment [4]. Chinese automaker BYD has developed BYD electric bus, all-electric bus model powered with its self-developed Iron-phosphate battery with the longest drive range of 250 km [82].

Frugal innovation is valuable to stimulate innovation in public services. Transportation of patience in rural areas of the developing countries is a daunting task. eRanger ambulance service for rural Africa and “dial 1298” in India use a cross-subsidy model to provide affordable or free services for poor patients [59,61]. Many small firms are developing supporting features for frugal transport. For example, a US firm Harman developed a system for emerging markets, “Saras” by using a simpler design and Indian and Chinese engineers [56]. Harman has managed to get Toyota as a customer [28]. IBM, for example, developed an algorithm to predict the volume of traffic in Nairobi based on data received from street webcams. Thus, it provides better traffic management in highly congested cities, such as Nairobi [48]. In the grassroots level, for example, farmers place water pumps in bullock carts to create a new mode of motorized transportation [73]. Thus, the transportation sector is evolving in various ways to meet the local and global needs.

4.8. Energy

Frugal innovation is considered as an important concept to provide energy to unserved customers. Solar energy, chargeable lights, and fuels for energy provision, such as rice husk bring opportunities for societal development. An Indian social enterprise SELCO provides solar-powered electricity, which is safer, greener, and cheaper than Kerosene-lit lamps or traditional sources of electricity, as these traditional sources emit harmful gases [4,71]. China's rapid and affordable production of solar panels are meeting the growing needs of the developing countries [6].

Starting in 2007, Husk Power, another social enterprise in India, has already lit up over 200 villages touching 200,000 lives [12,35]. Unlike other firms, this enterprise has formal training programs and willingness to franchise so that local entrepreneurs would operate systems built by the firm. Moreover, it is planning to expand in Southeast Asia and Africa [64]. Several other rice-husk based energy plants in other regions, such as Thailand [20,91] and China [99] are serving underserved customers. In Africa, Solar Sister aims to mitigate energy problem by empowering women with woman-centric sales networks [88].

Embracing frugal innovation in the energy sector can be useful for energy efficiency and materials sustainability [65]. Energy frugality is not just the only delivery of low-cost energy but also frugal consumption. Moreover, thinking alternatives of the existing low-cost energy consumption is essential. MittiCool, a fridge made of clay, does not need energy to keep food and vegetables fresh up to a week [78]. A holistic approach to frugal energy is emerging to include an increasing number of underserved customers.

4.9. Finance

Pioneered by Muhammad Yunus from Bangladesh, microfinance organization Grameen Bank has revolutionized the banking industry not only in poor countries but also in some Western countries [14]. The Grameen Foundation now has a footprint in 36 countries including in the Western world. Grupo Elektra, a Mexican financial and retail corporation, sends its loan officers to the homes of potential customers. Using handheld devices, the officers communicate the central office and provide an instant decision on banking services. Easily portable Vortex's frugal ATM solution for rural India consumes 90% less power than the standard ATM machines. It does not need air conditioning room even in very hot Indian weather. Vortex's ATM is run by solar power. Only five hours of sunshine is enough to charge the backup batteries for a day [70]. Moreover, its biometric system allows uneducated customers to make transactions without inserting numeric access code [43].

An Indian entrepreneur went a step ahead by combining smartphone technology with a biometric fingerprint scanner and creating an ATM on bicycle wheels [69]. The “missed call” service has been used in various sectors in India. In the banking sector, for example, a customer calls a bank number. Instead of answering the call, the bank calls back to the customer's number. Additionally, by an automated message, customers receive information, such as account balance and mini bank statement on their mobile device [70]. The Rickshaw Bank of the Center for Rural Development is empowering rickshaw pullers in some states of India [80]. In India, a company called A Little World offers a secured low-cost platform for financial services through special mobile devices that store and manage customer bank account using photo and biometric identification [78]. Thus, financial management is becoming simpler and making an impact on the lives of underprivileged people. Throughout the developing countries, as cited earlier in the ICT sector, Firms such as bKash, Easy paisa, Eko, Kopo Kopo and M-Pesa, and WIZZIT are actively serving their customers through mobile banking [12,13,94]. Thus, frugal finance is changing the lives of millions of underprivileged customers.

4.10. Water

Water is increasingly becoming scarcer, especially in the developing countries. Potable water and water conservation are essential part of innovation agenda [18]. Israel and the United Arab Emirates, the world's highest per capita consumer of water, are striving in desalinization [69]. They developed novel filtering systems to optimize individual units and reduce the plants' energy consumption. In 2008, Unilever's Indian unit launched a household water purifier Pureit that shows how low-income customers can stay safe from water-related diseases. Tata Chemicals' low-cost water purifier Tata Swach provides many customers the opportunity to drink pure water at an affordable cost [4]. Tata Swach has launched pricing at half the price of Pureit Classic in the Indian market in 2009 [77].

Technology on affordable water purification is advancing rapidly. For example, Swiss firm Vestergaard Frandsen developed a handheld water-purifying device called LifeStraw that can instantly purify water off bacteria and dirt up to 99.9% [90]. In India, Siemens' algae-bacteria based wastewater treatment system requires only 40% energy of a conventional sewage treatment plant [92]. Combined with portable water treatment and solar water heater system, a Swedish firm has developed a device called Solvatten to use at the household level in the developing countries. Solvatten is unique for the following reasons: easily portable, safe and warm water in 2–6 h, no need for chemicals and batteries, and it can be used several times a day [89]. Frugality is essential and more devices are necessary to manage water resource.

4.11. Miscellaneous

A Chinese electronics firm Haier captured over 60% market share in the USA in air conditioners market [4]. Haier has developed a series of frugal devices, such as Haier's fridge and Haier's washing machine. Another Chinses firm Galanz has been following the similar path [32,100]. An Indian conglomerate Godrej launched a portable, small-sized fridge called ChotuKool. Instead of a compressor, ChotuKool runs on thermoelectric cooling with a cooling chip. ChotuKool is operated by battery or inverter to keep its contents cool for up to three hours without a power connection [4,43]. Pepsico, for example, introduced lentils-based snacks in India, which found a place in the US market for reasons, such as healthy and low cost [4]. Exploring Unilever's washing-powder sachets, van Beers et al. [96] demonstrate how a Western multinational was successful with frugal innovation initially in Africa and then other parts of the world. The Hole in the Wall experiment on learning has already made an impact over 300,000 children in India and several African countries.

So far, the housing sector has received limited attention in frugal innovation studies with the exception of projects, such as 300$ House, Kutcha house for the poor in India and $5000 Awami Villas in Pakistan [84]. In the same vein, the agriculture sector is remained almost ignored in the literature. As a whole, frugal innovation is rapidly expanding in all sectors to empower the underserved customers and to narrow social segregation.

5. Implications, limitations and future research avenues

5.1. Implications

Adequate understanding of frugal innovation is essential for scholars, practitioners, and policy makers. Several implications have emerged from this study. Interestingly, most of the authors in the frugal innovation discipline are located in India. There are no articles published in top-tier (FT50) journals. The articles of frugal innovation have published in a wide range of disciplines with the highest concentration in Technology and Innovation category. Sector wise, healthcare holds a significant part of the frugal innovation literature as one-third of the cases are from that sector followed by cases related to electric and electronics sector. Most of the studies are conceptual and case studies. There are no quantitative studies on frugal innovation. Moreover, most of the case studies have used cases for illustration purpose. Hence, no studies adopted any rigorous analysis for building theory, model, or framework. Frugal innovation has high overlap with some emerging concepts, such as reverse innovation [33], disruptive innovation [38,100], grassroots innovation [86]. Many scholars have already attempted to delineate these overlapping of concepts [17,101].

Studies are emerging in a silo and making barriers in theory development. For example, many cases, such as Haier, Tata Nano, GE's ECG and Ultrasound machines are discussed in various disciplines. Apparently, almost any innovation that serves low-income customers is being increasingly considered as frugal innovation. It seems that some cases that have received high attention from the popular press, such as Jayashree Industries and Ksheera enterprises are missing or under-discussed in the literature. Some studies have not explored frugal innovation even though it is included as the keyword in the articles [87]. Studies point out that frugal innovations foster sustainability (see Refs. [17,57]). However, the degree of sustainability of frugal innovation has not been examined in a rigorous manner. The overwhelming majority of frugal innovations developed by grassroots innovators remained confined in local areas [1]. These innovations do not need significant finance.

Many MNCs of the developing countries are dethroning Western MNCs with their frugal innovations in the global market. Some Western firms are embracing frugal innovation passionately as a part of their business strategy. Thus, they are reaping a significant share of their revenues through frugal innovation. Moreover, the developing countries are highly growing markets and Western MNCs are localizing their business portfolios considering the local needs. However, some Western firms feel that embracing frugal innovation may cannibalize their market for high-end products. The advent of frugal innovations has blessed the healthcare sector for low-income customers. However, last-mile medication is still a far way to achieve [54].

Subsidiaries of Western MNCs in the developing countries need significant support from their Western counterparts to develop frugal devices [58]. Even though many MNCs from the developing countries are active in frugal innovation, they need strong support from Western firms both for sophisticated products, such as Tata Nano and for simple innovation, such as Jaipur foot (see e.g., Refs. [4,79]). Frugal innovations enhance inclusive innovation and thereby provide underprivileged people access to social and economic capital [54]. However, sectors such as agriculture, housing, water, and education have not been explored adequately. Solar energy and other popular energy solutions that are serving low-income customers considered as frugal innovation even though they existed at least for the last two decades.

5.2. Limitations

There are several limitations of this study. Documents such as masters and doctoral theses on frugal innovation have not included in this study. There might have some cases, which were not recorded from the literature even though a systematic searching approach was adopted. Moreover, anecdotal knowledge suggests that there are many more appropriate frugal innovation cases, which are missing in the literature. Some cases (e.g., LifeStraw and Solvatten) are barely mentioned so we need to search knowledge on those cases outside the frugal innovation discipline. Not all cases have been discussed here due to lack of information. The study conducted sectorial analysis whereas thematic analysis may bring different insights.

5.3. Future research directions

There are several avenues for future exploration. First, relating frugal innovation concept with other closely related concepts is essential to understand what frugal innovation is and what is not. As the boundary of frugal is not well established, a clear boundary and scope of frugal innovation concept are necessary. Second, studies have a high concentration on cases in India and Indian scholars share a significant proportion of authorship. Therefore, the contribution from the broad geographical contexts in terms of authorship and cases is necessary. Third, studies need to take rigorous research approach by moving from the existing conceptual and case-based research to deep investigation with quantitative research for theory building. Some regions (South America and Africa) and some countries (e.g., China) should get high attention to develop a holistic knowledge on frugal innovation. Sectors such as agriculture and education need to explore extensively as the study shows that these sectors have received limited attention in the literature even though anecdotal evidence indicate that there are ample of successful cases in these sectors. Fourth, top-tier scholars' engagement in this discipline may help to test the truth versus hype of frugal innovation. Fifth, this review study has adopted descriptive and sectorial analysis. A thematic analysis of the literature could be an appropriate approach to explore various themes that are present in the literature. Sixth, since frugal innovations need a different business model how large firms manage dual business models [97], one for costly products and another for frugal products is an interesting avenue for future research. Finally, sustainability is a key aspect of the business model, especially in the developing countries. Therefore, future studies may explore the sustainability of frugal innovation with appropriate measurement scales. Thus, frugal innovation may become a well-established discipline in the broad business literature.

Uncited references

[19], [26], [39], [40], [41], [42].

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Appendix 1. A list of 117 cases and their frequency of appearance in the frugal innovation literature

Case	#	Case	#	Case	#
Tata Nano	25	Galanz's microwave	1	CheX	1
GE's ECG machine	20	Haier's air conditioners	1	D-Rev	1

GE's Ultrasound machine	13	Haier's fridge	1	Diagnostics for paper-based diagnostic tests	1
Aravind Eye Care	10	Lenovo $300 laptop	1	Dry-Tri	1
Tata Swach	10	Zhongxing X-Ray Machine	1	Fetal Heart rate Monitor	1
Aakash	7	Nokia 1200	1	Fixed-dose combinations of Anti-retroviral drugs for AIDs	1
M-Pesa	7	Nokia basic phones	1	Food fortification with Iodine and Iron	1
ChutuKool	6	Siemens' Computed Tomography Scanner	1	Generic drugs	1
Husk Power Systems	6	Siemens' waste-water treatment	1	Harman's infotainment systems	1
MittiCool	6	"missed call" service in India	1	Health care	1
Narayana Hospital	6	A Little World-rural banking through mobile phony	1	HMI Line panel	1
Vortex ATMs	6	Acme power interface unit	1	Hole in the wall	1
GE's Lullaby baby warmer	5	ATM on wheels	1	HospIS project	1
Jaipur foot	5	Awaaz.De - Voice Message Board for Education	1	Invention Labs	1
Mahindra & Mahindra's small tractors	5	Bamboo Bike	1	Kutcha house	1
Embrace	5	Electronic Voting Machine (India)	1	Low cost sanitary toilets	1
Grameen Bank's micro-finance	4	Kerala Palliative Care	1	Low-cost bubble continuous positive airway pressure device (CPAP)	1
Easy Paisa	4	Motorcycle-based tractor	1	Mettler Toledo Weighing Scale	1
Haier's washing machine	3	Naandi - a system for clean water in Andhra Pradesh	1	Micro-PCR device	1
Logitech mouse	3	Scooter-powered flour-mill	1	Mindray- healthcare products	1
Bharti Airtel	3	Superseva	1	Missile	1
Rickshaw Bank in India	3	Kopo Kopo	1	Multix Select DR machine	1
SELCO	3	ToughStuff	1	Oral misoprostol	1
Nokia 1100	2	Medicall in Brazil	1	Oral rehydration therapy	1
Nokia's 101	2	300$ House- housing for the poor	1	OSDD	1
EKO mobile phone banking	2	bCPAP for new born breathing	1	Patient Monitoring System	1
Micromax's mobile phone (India)	2	Bahria Town	1	Ponseti method - gold standard treatment of club foot	1
Tata Ace	2	Dacia Logan	1	Probe for Detecting Tuberculosis	1
Unilever's Purit	2	Solvatten	1	Pureit	1
Vodafone Rs.10 pre-paid cellular phone balance	2	LifeStraw	1	Radio Telescope	1
Firefly - treat neonatal jaundice is under trial in Philippines and Vietnam	2	HTC $30 mobile	1	Reverse engineered vaccines	1
Solar Sister	2	Bamboo Microscope	1	Robotic Hand	1
BYD-Lithiumion Battery	2	Beating Heart Surgery	1	Small sachets of Tide for one rupee	1
Immunoassay-based fabric chips	2	WE CARE Solar Suitcase	1	Smart medicine pack by Microsoft Research India	1
Shakerscope	2	Well Baby Bassinet	1	Stove and Fuel	1
eRanger - ambulance for rural Africa	1	$5K Awami Villas	1	Super Religare Laboratories' Blood collection	1
Unilever's washing-powder sachets	1	Biofortification - to produce staple crops rich in micronutrients	1	Vivian Fonseca-SMS message to control diabetes	1
Mobile e-learning in Bangladesh	1	Bone Drill	1	WiMax	1
Dachangjiang- Motorcycle	1	careHPV	1	XCyto Screen series	1

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